
L2 Regularization Reliably Reduces Overfitting but Does Not Consistently Improve Accuracy on Small Tabular Classification Datasets

example_user and NeuriCo

Abstract

Small tabular datasets are a regime where practitioners regularly reach for L2 regularization, but the size and consistency of its benefit are often unclear. We study a simple, controlled version of that question: does validation-tuned L2 regularization improve held-out generalization relative to an otherwise matched logistic-regression baseline on datasets with fewer than 1000 samples? We evaluate four real classification datasets with stratified 70/15/15 train/validation/test splits and three fixed random seeds per dataset. For each split, we compare a near-unregularized logistic regression baseline, a validation-tuned L2 logistic regression model, and a stratified dummy classifier. We report test accuracy, weighted F1, train-validation and train-test gaps, coefficient norms, and paired statistical tests over the 12 dataset-seed pairs.

L2 regularization improves the mean test accuracy from 0.8267 to 0.8385, but that gain is not statistically significant ($p = 0.205$). The same pattern holds for weighted F1 (0.8257 to 0.8357, $p = 0.302$). In contrast, L2 sharply reduces the mean train-validation gap from 0.1297 to 0.0257, a paired reduction of 0.1040 with a 95% confidence interval of $[-0.1792, -0.0287]$ and $p = 0.011$. This gap shrinks in 11/12 runs, while test accuracy improves in only 5/12 runs, ties in 5/12 runs, and worsens in 2/12 runs.

Our main conclusion is practical rather than universal: on small real tabular datasets, L2 regularization is a dependable anti-overfitting tool, but not a reliable accuracy booster. That distinction matters when regularization is used to justify stronger claims about predictive performance.

1 Introduction

Small tabular classification problems are common in applied machine learning, and they are exactly the setting where overfitting can dominate model selection. When a dataset has only a few hundred samples, a single validation split can make a method look either robust or unstable depending on the random seed. L2 regularization is often treated as a safe default in this regime, yet the practical question is narrower than that habit suggests: does L2 only shrink models, or does it also improve held-out performance in a consistent way?

Classical theory motivates optimism. Weight decay prefers smaller-norm solutions and can suppress variance and noise fitting Krogh and Hertz [1991]. Modern work, however, has made that story more conditional. The effect of L2 depends on optimization details Loshchilov and Hutter [2019], and explicit regularization alone does not explain generalization in modern models Zhang et al. [2017]. Recent papers also argue that regularization changes optimization dynamics and can be especially relevant in small-data or overtraining-prone settings Power et al. [2022], D’Angelo et al. [2024], Golatkar et al. [2019]. What is missing is a compact, repeated-seed measurement study focused on tiny real tabular datasets rather than vision benchmarks or synthetic tasks.

We address that gap with a controlled benchmark on four classification datasets with fewer than 1000 samples. Our setup keeps the model family, preprocessing, solver, and split protocol fixed, and changes only the regularization strength. We compare a near-unregularized logistic regression baseline against a validation-tuned L2 logistic regression model, evaluate both on untouched test splits, and report both predictive metrics and direct overfitting indicators. This framing is intentionally modest: we do not ask whether L2 solves generalization in the abstract; we ask how much it changes accuracy, weighted F1, and train-validation gaps in a small-data tabular workflow.

The answer is mixed. Across 12 dataset-seed pairs, L2 increases mean test accuracy by only 0.0118 and weighted F1 by 0.0100, with confidence intervals that cross zero. At the same time, it reduces the mean train-validation gap by 0.1040 and does so in 11/12 runs. This means the overfitting-control effect is strong and repeatable, while the accuracy effect is weaker and dataset-dependent.

What do we contribute? We make four concrete contributions:

- We conduct a repeated-seed comparison of near-unregularized and validation-tuned L2 logistic regression on four real small-data tabular benchmarks.
- We separate predictive performance from overfitting control by reporting accuracy, weighted F1, train-validation gaps, train-test gaps, and coefficient norms.
- We quantify aggregate and per-dataset effects with paired t -tests, confidence intervals, and paired Cohen’s d rather than relying on point estimates alone.
- We provide a reproducible artifact-driven paper with figures, tables, and exact environment details for the measured setting.

The rest of the paper is organized as follows. Section 2 positions our study in the L2 regularization literature. Section 3 describes the experimental design. Section 4 presents the aggregate and per-dataset findings, and Section 5 interprets their implications and limitations before Section 6 concludes.

2 Related Work

Classical weight decay. The classical argument for L2 regularization is that smaller-norm solutions can improve generalization by reducing variance and limiting sensitivity to target noise Krogh and Hertz [1991]. That view motivates our core expectation that L2 should shrink coefficients and reduce the train-validation gap on small datasets. Our study directly tests that practical implication in a simple linear model rather than in a high-capacity deep network.

Optimizer semantics and implementation details. Later work showed that L2 regularization and weight decay are not always interchangeable. Loshchilov and Hutter Loshchilov and Hutter [2019] demonstrate that the equivalence breaks under adaptive optimizers, which matters whenever a comparison loosely uses “weight decay” to mean several different update rules. Our setting avoids that ambiguity by using the same lbfgs logistic-regression solver in both conditions and changing only the penalty strength.

Regularization is not the whole story. Zhang et al. Zhang et al. [2017] caution against explanations of generalization that rely too heavily on explicit regularizers. That caution is directly relevant here: even if L2 reduces overfitting indicators, it need not deliver a reliable test-accuracy gain. We therefore treat predictive improvement and gap reduction as separate empirical questions instead of assuming they move together.

Small-data and dynamics-oriented views. Recent work strengthens the case for studying regularization in low-data regimes while still keeping the claim conditional. Power et al. Power et al. [2022] show that small datasets can exhibit long delays between memorization and generalization, with regularization affecting data efficiency. Golatkar et al. Golatkar et al. [2019] argue that the timing of regularization matters, especially early in training. D’Angelo et al. D’Angelo et al. [2024] further reinterpret weight decay as a mechanism that shapes optimization dynamics rather than only limiting capacity. These papers motivate our small-data focus, but they also suggest that predictive gains may depend strongly on dataset structure and training conditions.

Position of this work. Unlike prior deep-learning studies, we focus on a narrow but common practical setting: tiny real tabular classification datasets paired with logistic regression. Unlike purely theoretical treatments, we quantify both the size and the consistency of the observed L2

Table 1: Datasets used in the study. All datasets have fewer than 1000 samples.

Dataset	Samples	Features	Classes	Task
WINE	178	13	3	multiclass
BREAST CANCER	569	30	2	binary
GLASS	214	9	6	multiclass
SONAR	208	60	2	binary

effect across repeated stratified splits. Our contribution is therefore empirical and measurement-oriented: we document when shrinkage reliably appears and when predictive gains fail to rise above split noise.

3 Methodology

Study design. Our main question is whether L2 regularization improves held-out generalization on small classification datasets compared with an otherwise matched logistic-regression baseline. For each dataset and seed, we create one stratified 70/15/15 train/validation/test split, fit preprocessing on the training partition only, tune regularization strength on the validation partition, and evaluate once on the untouched test split.

Datasets. We study four local CSV datasets drawn from `datasets/`: WINE (178 samples, 13 features, 3 classes), BREAST CANCER (569 samples, 30 features, 2 classes), GLASS (214 samples, 9 features, 6 classes), and SONAR (208 samples, 60 features, 2 classes). The exploratory data analysis artifact reports no missing values and no duplicate rows in these four primary datasets. Because BREAST CANCER, GLASS, and SONAR show moderate class imbalance, we report weighted F1 alongside accuracy.

Preprocessing and models. We encode labels with `LabelEncoder` and standardize features with `StandardScaler`. The scaler is fit on the training split only and then applied to validation and test splits to avoid leakage. We compare three models:

- STRATIFIED DUMMY: `DummyClassifier(strategy="stratified")` as a sanity lower bound.
- NEAR-UNREGULARIZED: logistic regression with a near-unregularized setting, implemented as `C=1e12`. This is an implementation workaround for the `scikit-learn 1.8.0` API rather than an exact zero-penalty solver.
- L2: logistic regression with L2 regularization and validation-based hyperparameter selection over a logarithmic grid of $C \in \{10^{-4}, 3 \cdot 10^{-4}, \dots, 10^3\}$.

All logistic-regression runs use the `lbfgs` solver and `max_iter=5000`.

Selection metric and evaluation metrics. We choose the best L2 strength by validation weighted F1, then report test accuracy, weighted F1, train-validation accuracy gap, train-test accuracy gap, and coefficient ℓ_2 norm. We focus especially on the train-validation gap because it gives a direct measure of overfitting under the same split protocol.

Statistical analysis. We compare L2 against NEAR-UNREGULARIZED using paired `ttest_rel` across the 12 dataset-seed pairs. For each aggregate metric, we report the mean paired difference, a 95% confidence interval, paired Cohen’s d , and a Shapiro–Wilk test on the paired differences as a descriptive assumption check. We treat per-dataset tests as descriptive because each one uses only three seeds.

Implementation details. The experiment runs on CPU only and records Python 3.12.8, NumPy 2.4.4, pandas 3.0.2, scikit-learn 1.8.0, SciPy 1.17.1, Matplotlib 3.10.8, and seaborn 0.13.2. The latest validation execution takes about 8 seconds. Table 1 summarizes the datasets, and the main quantitative comparisons appear in table 2 and table 3.

Metric	Baseline Mean	L2 Mean	Mean Diff. (Δ)	95% CI for Δ	p -value	Paired d	Better
Test accuracy	0.8267	0.8385	+0.0118 $\uparrow$	$[-0.0075, 0.0311]$	0.205	0.389	L2
Weighted F1	0.8257	0.8357	+0.0100 $\uparrow$	$[-0.0103, 0.0303]$	0.302	0.312	L2
Train-val gap	0.1297	0.0257	-0.1040 $\downarrow$	$[-0.1792, -0.0287]$	0.011	-0.878	L2

Table 2: Aggregate paired comparison across 12 dataset-seed pairs. Higher is better for accuracy and weighted F1, while lower is better for the train-validation gap. Best values are shown in **bold**. Only the gap reduction is statistically significant at $\alpha = 0.05$.

Dataset	Dummy Acc.	Baseline Acc.	L2 Acc.	Baseline F1	L2 F1	Baseline Gap	L2 Gap
BREAST CANCER	0.558	0.946	0.977	0.946	0.977	0.027	-0.011
WINE	0.321	0.963	1.000	0.963	1.000	0.049	0.019
SONAR	0.448	0.802	0.781	0.802	0.780	0.301	0.049
GLASS	0.273	0.596	0.596	0.592	0.586	0.141	0.046

Table 3: Per-dataset mean results across three seeds. Higher is better for accuracy and weighted F1, and lower is better for the train-validation gap. The pattern is mixed for predictive metrics but consistently favors L2 for gap reduction. Median selected C values are 0.10 for BREAST CANCER, 0.01 for WINE, 0.10 for SONAR, and 100.0 for GLASS.

4 Results

Aggregate comparison. Table 2 shows the main aggregate outcome. L2 increases mean test accuracy from 0.8267 to 0.8385, but the mean paired gain is only +0.0118 with a 95% confidence interval of $[-0.0075, 0.0311]$ and $p = 0.205$. Weighted F1 shows the same pattern: a small mean gain of +0.0100 with $p = 0.302$. In contrast, the train-validation gap drops from 0.1297 to 0.0257, a mean paired change of -0.1040 with $p = 0.011$ and paired Cohen’s $d = -0.878$. This is the clearest statistically supported effect in the study.

Per-dataset behavior. Table 3 shows that the aggregate average hides substantial dataset heterogeneity. BREAST CANCER and WINE both favor L2 on accuracy and weighted F1, with WINE reaching perfect mean test accuracy under the selected settings. SONAR and GLASS show the opposite pattern: L2 shrinks the train-validation gap strongly but does not improve mean test accuracy. This split is important because it shows that reduced overfitting pressure does not automatically imply better test performance.

How consistent is the overfitting reduction? The run-level counts sharpen the interpretation. L2 improves test accuracy in 5/12 runs, matches the baseline in 5/12, and worsens it in 2/12. By contrast, it reduces the train-validation gap in 11/12 runs. The regularization effect is therefore far more stable for shrinkage and gap control than for final predictive accuracy.

Regularization sweep and shrinkage. Figure 1 summarizes the validation sweep. The wine example shows that intermediate values of C can reach perfect validation and test performance while keeping coefficients much smaller than the near-unregularized baseline. Across datasets, the selected median C values vary from 0.01 on WINE to 100 on GLASS, which reinforces that the best amount of regularization is data dependent rather than universal.

Distribution of train-validation gaps. Figure 2 shows the gap distribution by dataset. The largest contraction appears on SONAR, where the mean gap drops from 0.3011 to 0.0486. GLASS also shows a large reduction, from 0.1408 to 0.0455. The visual pattern matches the paired test: L2 acts as a strong anti-overfitting mechanism even when accuracy gains are weak or absent.

Sensitivity analysis. The L2 sweep also serves as a simple ablation on regularization strength. Extremely small C values underfit, as expected, while very large C values recover the high-norm near-unregularized regime. The useful region lies in between, but that region differs by dataset. This is consistent with our main conclusion: regularization helps most as a controllable bias-variance tradeoff, not as a uniform test-accuracy improvement.

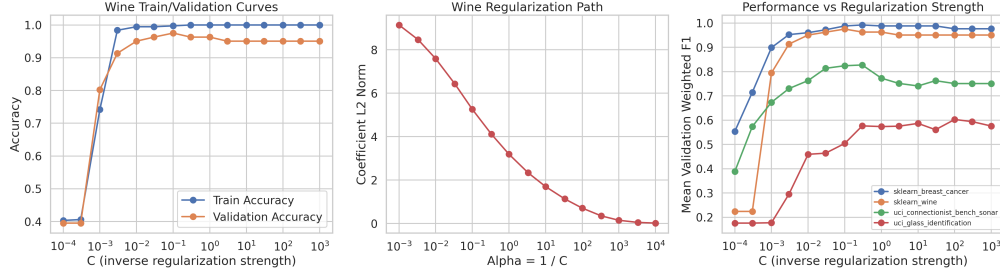


Figure 1: Regularization summary figure from the experiment artifacts. (Left) shows train and validation curves on WINE across the L2 sweep. (Center) shows coefficient shrinkage along the same path. (Right) aggregates validation weighted F1 as a function of C across datasets. The figure highlights the central pattern of the paper: performance can improve at intermediate regularization strengths, but the location and size of the gain vary by dataset.

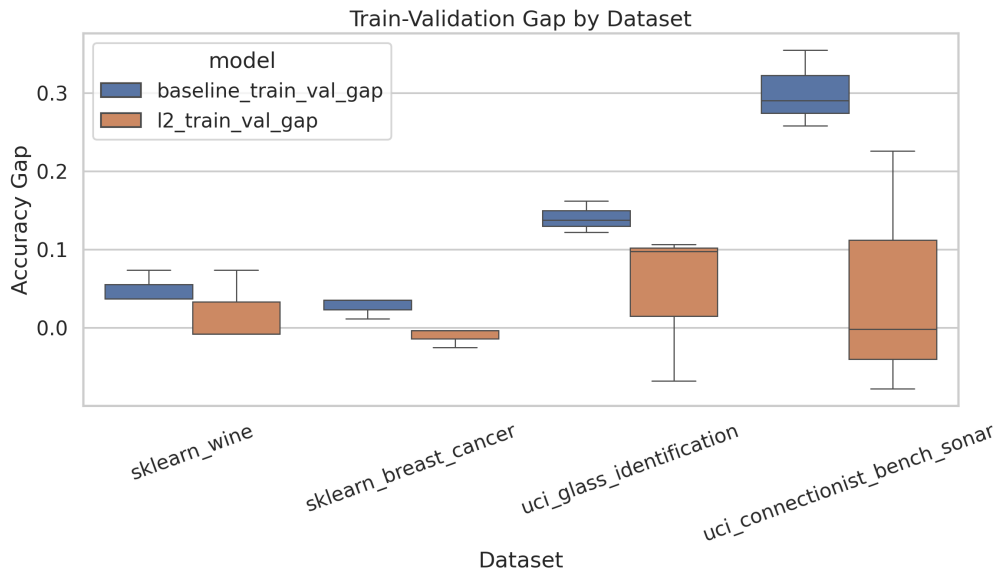


Figure 2: Train-validation accuracy gap by dataset for the near-unregularized baseline and the validation-tuned L2 model. Lower is better. L2 reduces the gap for nearly every run and produces the clearest effect on the harder small datasets, especially SONAR and GLASS.

5 Discussion

What is the strongest claim supported by the data? The strongest claim is that L2 regularization reliably reduces overfitting in this small-data setting. The average train-validation gap falls by about 0.104, the paired confidence interval excludes zero, and the effect appears in 11/12 runs. That is exactly the pattern one would expect if shrinkage were acting mainly as variance control.

Why is the accuracy effect weaker? Accuracy combines both variance reduction and bias increase. On easier datasets such as BREAST CANCER and WINE, shrinkage reduces variance without paying much cost in bias, so both predictive performance and gap metrics improve. On more difficult datasets such as SONAR and GLASS, the same shrinkage can remove enough flexibility to offset any variance benefit. The result is a smaller gap but no reliable accuracy gain. This interpretation is aligned with the conditional view of regularization in prior work Zhang et al. [2017], Power et al. [2022], D’Angelo et al. [2024].

What does this imply for practice? For small tabular datasets, L2 is well justified when the goal is to stabilize a linear model and control apparent overfitting. It is less justified if the claim is that it will reliably increase held-out accuracy. Practitioners should therefore treat L2 as a dependable

shrinkage tool and tune it carefully, rather than assuming it is a universal improvement over a weakly regularized baseline.

Limitations. This study has four important limitations. First, it only evaluates logistic regression, so the results may not transfer directly to multilayer perceptrons or tree-based models. Second, the aggregate analysis covers only 12 paired observations, and the per-dataset analyses use only three seeds each. Third, the baseline is only approximately unregularized because it uses $C=1e12$ for compatibility with the current `scikit-learn` interface. Fourth, the split protocol uses a single validation split per seed rather than nested cross-validation, so selection variance may still be understated. In addition, the Shapiro–Wilk test rejects normality for the aggregate train-gap differences, which suggests caution when reading the corresponding paired t -test even though the effect size is large.

Broader implications. The broader lesson is methodological. In small-sample settings, it is easy to overstate a regularization result if one reports only test accuracy on one split. Our results show why paired repeated-split analysis matters: the same intervention can have a strong and statistically supported effect on overfitting indicators while failing to deliver a statistically supported gain on final accuracy.

6 Conclusion

We studied whether validation-tuned L2 regularization improves held-out generalization for small tabular classification datasets relative to an otherwise matched logistic-regression baseline. Across four datasets and three random seeds per dataset, L2 produced a clear and statistically supported reduction in the train-validation gap, but only a small and non-significant increase in aggregate test accuracy and weighted F1.

The key takeaway is simple: in this setting, L2 regularization is a reliable anti-overfitting mechanism, not a universal accuracy booster. That is a useful distinction for practitioners and for empirical papers that interpret regularization effects too broadly.

Several extensions would strengthen the picture. Future work should repeat the protocol for small multilayer perceptrons, replace single validation splits with repeated nested cross-validation, add calibration and confusion-matrix analysis, and test whether the same pattern persists on a larger collection of small tabular datasets or stratified subsamples of larger benchmarks.

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